

Ricardo Valero

Software Engineer

I build typed, declarative systems — effects that declare their requirements and failures, schemas at every boundary, builds that reproduce anywhere — and I keep the tooling fast and light.

ricardo-valero ricardo-valero-de-la-rosa

WORK EXPERIENCE

Manuable, S.A.P.I. de C.V.

Senior Software Engineer

04.2025 – Present

Multi-provider Shipping Platform

- Hardened provider integrations (FedEx, DHL, Estafeta): rate pricing and margins, label recovery flows, and error surfacing
- Built and refined UI components across the product and fixed N+1 query performance issues

Ruby Rails PostgreSQL JavaScript

Integration Hub — Provider Shipment Service

- Designed and built solo a standalone service executing provider ship-quote/rate and label-generation calls for multiple consumer applications
- Stateless on tenancy: provider credentials exchanged for encrypted session bearers, never persisted at rest

Elixir Phoenix

Data Hub — Centralized Ingestion and Transformation

- Built ingestion pipelines landing source systems into BigQuery with a SQLMesh transformation layer, as a data-stack monorepo
- Consolidated earlier fragmented ETL services (Airbyte, ad-hoc GCP jobs) into a single platform

Python dlt SQLMesh BigQuery GCP Metabase

Invoice Data Ingestion — Spreadsheets to BigQuery

- Transformed raw Excel, CSV, and spreadsheet invoice data into structured BigQuery ingestions with multi-sheet imports

Elixir Phoenix libSQL BigQuery

Infrastructure Tooling

- Built a Nix-managed CLI deploying a self-hosted PaaS (Dokploy) and compose services across AWS and GCP
- Fleet management for company services — machines, DNS, secrets, and deploys — one VM per app

Nix NixOS Terraform AWS GCP Docker

Hitower Technohub, S. de R.L. de C.V.

Senior Software Engineer

09.2024 – 02.2025

SEO and Content Optimization Web Service for Improved Search Rankings (Write2Rank)

- Developed a full-stack web application for content analysis and ranking suggestions.
- Designed and implemented scalable backend APIs and frontend interfaces.

PHP Laravel PostgreSQL TypeScript Node.js Next.js

CI/CD Pipeline Optimizations

- Conducted experiments to improve build times and deployment processes.
- Implemented reproducible builds to enhance reliability.
- Optimized caching and container usage for faster CI/CD.

AWS GitLab CI Docker Nix

Diseño Control y Sistemas a la Medida, S.A. de C.V. (aka Industrial Code)

Senior Software Engineer

04.2021 – 05.2023

Assembly Line Real-time Production Monitoring Web Application for Polaris Inc.

- Led design and architecture of UI/UX, tech stack, database, and APIs

TypeScript Node.js Fastify tRPC DrizzleORM PostgreSQL React

Vehicle Registration Hybrid Web and Mobile Application (Registegic)

- Led design and architecture of UI/UX, tech stack, database, and APIs

AWS TypeScript Node.js Fastify tRPC Prisma PostgreSQL React

Warehouse Repackaging Control Web Application for Polaris Inc.

- Led design and architecture of UI/UX, tech stack, database, and APIs

TypeScript Node.js Fastify GraphQL Prisma PostgreSQL React

Hardware Integration for IoT

- Designed tech stack and APIs for serial adapters, printers, tools, and IoT sensors

TypeScript Node.js SQLite

Tecnología en Sistemas de Integración S.A. de C.V. (TSI)

Junior Software Engineer

01.2019 – 02.2021

Assembly Line Control Applications for Polaris Inc.

- Developed custom SCADA/MES, hardware integration, data collection, database design, and APIs

VBScript Indusoft REST OpenProtocol ODBC

Work Shift Management Web Application for Polaris Inc.

- Developed a full-stack app, database design, and APIs

C# .NET REST EntityFramework MSSQL React

Assembly Line Reporting Web Application for Polaris Inc.

- Developed a full-stack app, database design, and APIs

C# .NET REST EntityFramework MSSQL Razor

Internship in Automation Engineering

08.2018 – 12.2018

Ladder-logic PLC Program Development

- In charge of several ladder-logic PLC programs for industrial clients, on Honeywell and Rockwell controllers

Ladder Logic Honeywell PLC Rockwell PLC

Institut für Mikrostrukturtechnik (IMT), Karlsruher Institut für Technologie (KIT)

Internship in Advanced Materials and Optical Spectroscopy

03.2017 – 07.2017

Automated Dip-coating Control for SURMOF Deposition

- Programmed MATLAB timing control for a dip-coating machine in solar-cell research: immersion, hold, withdrawal, and transfer between liquid baths
- Optimized thin-film deposition for large-area SURMOF layers

MATLAB

SIDE PROJECTS

Deplugger — Type-safe Workflow Platform

- Designed an op/workflow DSL where workflows are data: every op declares input/output schemas, so composed workflows are validated end-to-end at every step
- Built a dual-target compiler: the same workflow runs as an ephemeral effect or compiles to a durable, replay-safe workflow with activity semantics
- Streaming RPC surfaces live run telemetry to the UI; credentials encrypted at rest with schema-derived secret redaction

TypeScript Effect Bun SolidJS SQLite Nix Cloudflare

Raggio — Schema Library for Elixir

- Port of Effect-TS Schema to Elixir: composable, bidirectional schemas for decoding, encoding, and validation with typed errors

Elixir

EDUCATION

Universidad Autónoma de Nuevo León (UANL) — Facultad de Ingeniería Mecánica y Eléctrica (FIME)

08.2013 – 12.2018

Bachelor in Mechanical and Electrical Engineering
Specialization in Thermodynamics and Refrigeration

Karlsruher Institut für Technologie (KIT)

08.2016 – 07.2017

Academic Exchange in Mechanical Engineering
Scholarship from the German Academic Exchange Service (DAAD)

SKILLS

Languages	TypeScript, Python, Ruby, Elixir, C#, PHP, SQL
Backend	Node.js, Bun, Deno, Express, Fastify, tRPC, Effect, DrizzleORM, Prisma, Rails, Phoenix, .NET, EntityFramework, Laravel
Data	dlt, SQLMesh, BigQuery, Metabase
Frontend	Next.js, React, SolidJS, Svelte, WebAPIs, WASM
DevOps	AWS, GCP, Docker, Nix, Terraform, GitLab CI, GitHub Actions
Databases	PostgreSQL, MariaDB, MSSQL, SQLite
Protocols	SSH, HTTP, WebSockets, REST, GraphQL, gRPC, OpenProtocol, ODBC
Daily Drivers	macOS, Ghostty, Zed, Claude Code, opencode, Nix, direnv, OpenSpec

LANGUAGES

Spanish	native
English	advanced
German	basic